

Experiment No. 4

Title:

Use Function Types, Scope, and Closures; Apply Try-Catch; Build a Palindrome Checker

Software/Tools Required

- Visual Studio Code (or any text editor)
- Web Browser (Google Chrome, Microsoft Edge, Firefox, etc.)

Theory

1. Function Types in JavaScript

Functions are reusable blocks of code that perform a specific task.

a) Function Declaration

A named function declared using the `function` keyword.

```
function greet() {  
  console.log("Hello");  
}
```

Advantages

- Can be called before its declaration.
- Easy to read and maintain.

b) Function Expression

A function assigned to a variable.

```
const greet = function() {  
  console.log("Hello");  
};
```

Advantages

- Useful for callbacks.
- Can be passed as arguments.

c) Arrow Function (ES6)

Provides a shorter syntax for writing functions.

```
const greet = () => {  
  console.log("Hello");  
};
```

Advantages

- Short and concise syntax.
- Does not have its own `this` value.

2. Scope

Scope determines where variables can be accessed.

Global Scope

A variable declared outside any function is accessible throughout the program.

```
let name = "Radhika";
```

Local Scope

Variables declared inside a function can only be used within that function.

```
function demo() {  
  let age = 20;  
}
```

Block Scope

Variables declared using `let` or `const` inside `{ }` are accessible only within that block.

```
if(true){  
  let x = 10;  
}
```

3. Closures

A **closure** is a function that remembers and can access variables from its outer function even after the outer function has finished executing.

Applications

- Data hiding
- Private variables
- Function factories

4. Exception Handling (Try-Catch)

JavaScript uses **try-catch** blocks to handle runtime errors without stopping the execution of the program.

Syntax

```
try{
  // Code
}
catch(error){
  // Error handling
}
```

Advantages

- Prevents program crashes.
- Improves reliability.
- Displays meaningful error messages.

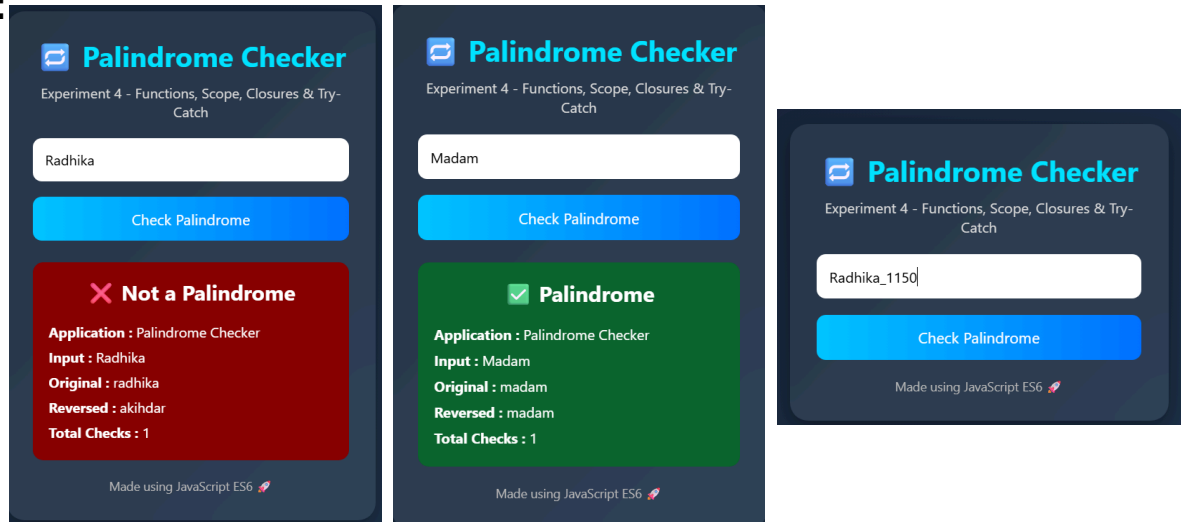
5. Palindrome Checker

A **palindrome** is a word, number, or phrase that reads the same forward and backward after removing spaces and converting to lowercase.

Working of the Program

1. Accept input from the user.
2. Convert the string to lowercase.
3. Remove spaces (optional).
4. Reverse the string.
5. Compare the original and reversed strings.
6. If both are equal, display **Palindrome**.
7. Otherwise, display **Not a Palindrome**.
8. Use **try-catch** to handle invalid or empty input.

Output:



Result/Conclusion:

The experiment was successfully completed. Different types of JavaScript functions, variable scopes, closures, and exception handling using **try-catch** were studied and implemented. A **Palindrome Checker** was developed that accepts user input, validates it using exception handling, and correctly determines whether the entered string is a palindrome. This experiment enhanced the understanding of JavaScript functions, scope management, closures, and error handling in real-world applications.